

SIMATS ENGINEERING

ASSESSMENT TOOL II — Visualization Analysis Report

Course: Data Handling and Visualization	Code: DSA06	CO: CO5
Duration: 180 min	Total Marks: 100	Reg No : 192524108 Name:S. LAKSHITA

COURSE OUTCOME COVERED

CO5: Analyze and evaluate visualization choices to communicate patterns, comparisons, relationships, and potential misleading effects through appropriate visualization methods. (BL4)

Activity Tool : Visualization Analysis Report

Weightage: 20%

Code of Conduct:

I S.LAKSHITA(192524108) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Assessment Criteria

Criteria	Visualization Selection and Understanding 25%	Visualization Analysis and Interpretation 25%	Application of Visualization Principles 20%	Organization and Presentation 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Visualization Analysis Report – Assessment Tool**R Programming Practical Questions with Datasets**

Instructions: Answer all five questions. Use R programming to construct the required visualizations, analyze the patterns, compare visualization choices where requested, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

Q1. Comparing Distributions – Histogram, Density and ECDF

A quality-control team records the following processing times (in minutes) for 20 randomly selected tasks. Using R, create a histogram and a density plot for the data. Also construct an empirical cumulative distribution function (ECDF). Compare the three visualizations and explain which aspects of the distribution—center, spread, skewness, and cumulative behavior—are communicated most clearly by each.

Data: Time = 18, 21, 22, 24, 25, 25, 27, 28, 29, 30, 31, 32, 33, 35, 36, 38, 41, 43, 46, 50 *Expected R tasks: histogram, density estimation, ECDF, appropriate labels/scales, comparison and interpretation.*

Analysis

A researcher measures the following response times (in seconds). Use R to create a normal Q-Q plot and a histogram/density visualization. Assess whether the observations are reasonably consistent with a normal distribution. Identify any departures from the reference line and explain what they indicate about the distribution.

Data: Response = 42, 45, 47, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 60, 61, 63, 66, 70, 76, 85

Expected R tasks: qqnorm()/qqline() or ggplot2 Q-Q plot, histogram/density plot, interpretation of deviations and outliers.

Q3. Visualizing Many Distributions at Once

Three groups of students have examination scores recorded below. Create a suitable visualization that compares the distributions of all three groups at once. Use boxplots and/or violin plots, and optionally overlay individual observations. Analyze differences in center, spread, and possible outliers. Explain why the selected visualization is more informative than displaying three separate plots.

Group A = 52,55,57,58,60,61,62,64,65,66,68,70,72,74,78;

Group B = 45,49,51,53,55,56,58,59,60,62,63,65,67,69,71;

Group C = 58,60,61,63,64,65,67,68,70,71,72,74,76,80,84

Expected R tasks: data reshaping, grouped boxplot/violin plot, optional jitter, comparison of distributions, interpretation.

Q4. Relationships and Multivariate Visualization

A study records advertising expenditure, website visits, and sales for 15 observations. Create a scatterplot of advertising expenditure versus sales and use an appropriate visual encoding to incorporate website visits. Add a fitted trend line. Analyze the strength and direction of the relationship and explain whether the additional variable improves the interpretation or risks cluttering the visualization.

Advertising = 10,12,14,16,18,20,22,24,26,28,30,32,34,36,38;

Visits = 120,135,150,165,180,195,210,225,245,260,275,290,305,320,340;

Sales = 48,50,53,55,58,61,63,66,69,71,74,77,80,83,86

Expected R tasks: scatterplot, trend line, third-variable encoding, appropriate legend/scale, relationship analysis and critique.

Q5. Misleading Visualization and Redesign

A college reports the following pass percentages for seven departments. Create two visualizations: (1) a misleading version using a truncated y-axis and an inappropriate visual emphasis, and (2) a redesigned version using an appropriate baseline, ordered categories, and clear labels. Compare the two plots and explain how the first visualization can exaggerate differences and why the redesigned visualization communicates the data more honestly.

Department = CSE, ECE, EEE, MECH, CIVIL, AIDS, IT;
Pass_Percentage = 91,93,92,89,90,95,94

Expected R tasks: bar chart with controlled y-axis, redesigned bar chart, ordering, labels, misleading-axis critique, accessibility and communication analysis.

CO5 AT2 – Visualization Analysis Report Answers

Based directly on the uploaded question paper.

Q1. Comparing Distributions – Histogram, Density and ECDF

Aim

To visualize and compare the distribution of processing times using a histogram, density plot, and ECDF.

R Code

```
Time <- c(18,21,22,24,25,25,27,28,29,30,31,32,33,35,36,38,41,43,46,50)
```

```
# Histogram
```

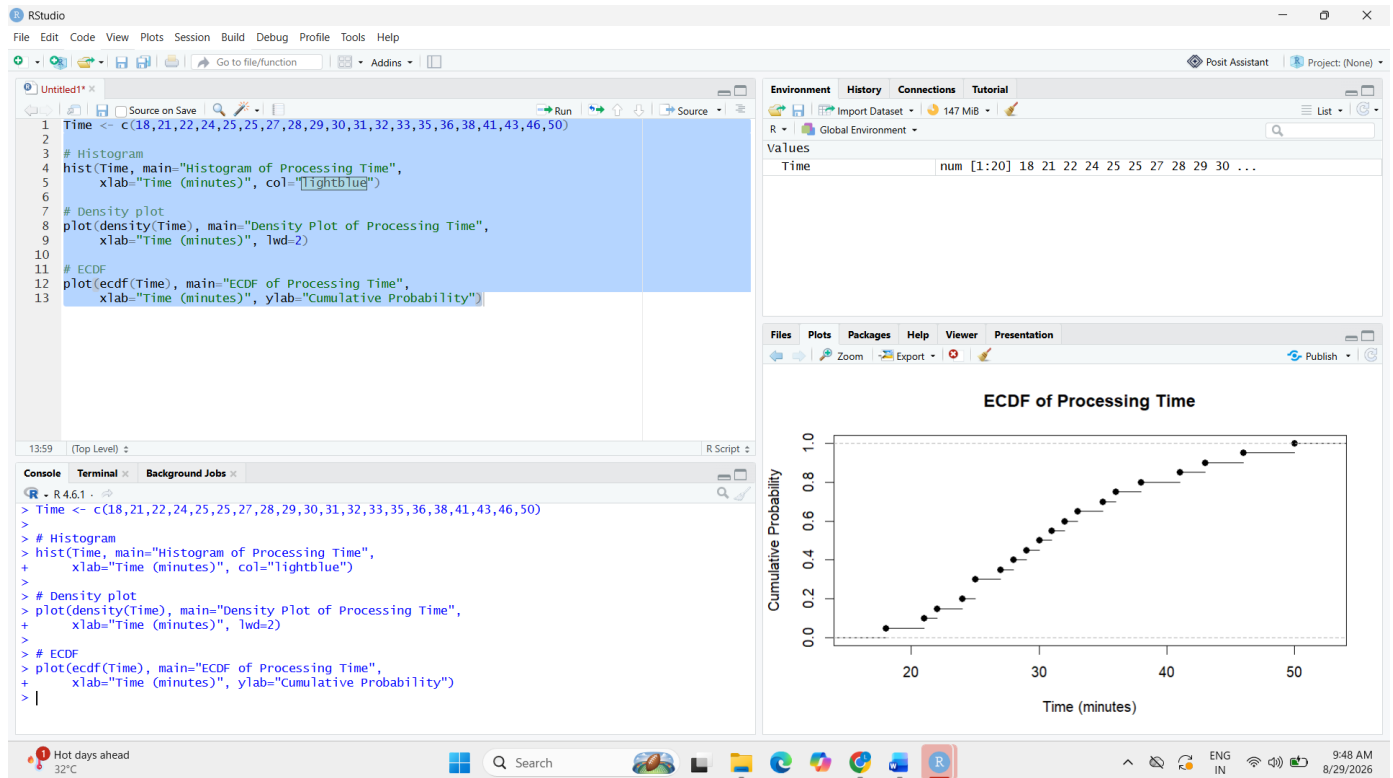
```
hist(Time, main="Histogram of Processing Time",  
      xlab="Time (minutes)", col="lightblue")
```

```
# Density plot
```

```
plot(density(Time), main="Density Plot of Processing Time",  
      xlab="Time (minutes)", lwd=2)
```

```
# ECDF
```

```
plot(ecdf(Time), main="ECDF of Processing Time",  
      xlab="Time (minutes)", ylab="Cumulative Probability")
```



Result / Interpretation

- **Histogram:** Shows the frequency distribution and overall shape.
- **Density plot:** Gives a smooth view of the distribution and clearly shows the center and spread.
- **ECDF:** Shows the cumulative proportion of observations below each time.
- Processing times are centered around **30–35 minutes**.
- The distribution has a **slight right skew** because of higher values such as 43, 46 and 50.
- ECDF is best for understanding cumulative behavior.

Q2. Q-Q Plot and Normality Analysis

Aim

To examine whether response times approximately follow a normal distribution using a Q-Q plot and histogram/density plot.

R Code

```

Response <- c(42,45,47,49,50,51,52,53,54,55,
              56,57,58,60,61,63,66,70,76,85)

```

```
# Q-Q Plot
```

```
qqnorm(Response, main="Normal Q-Q Plot")
```

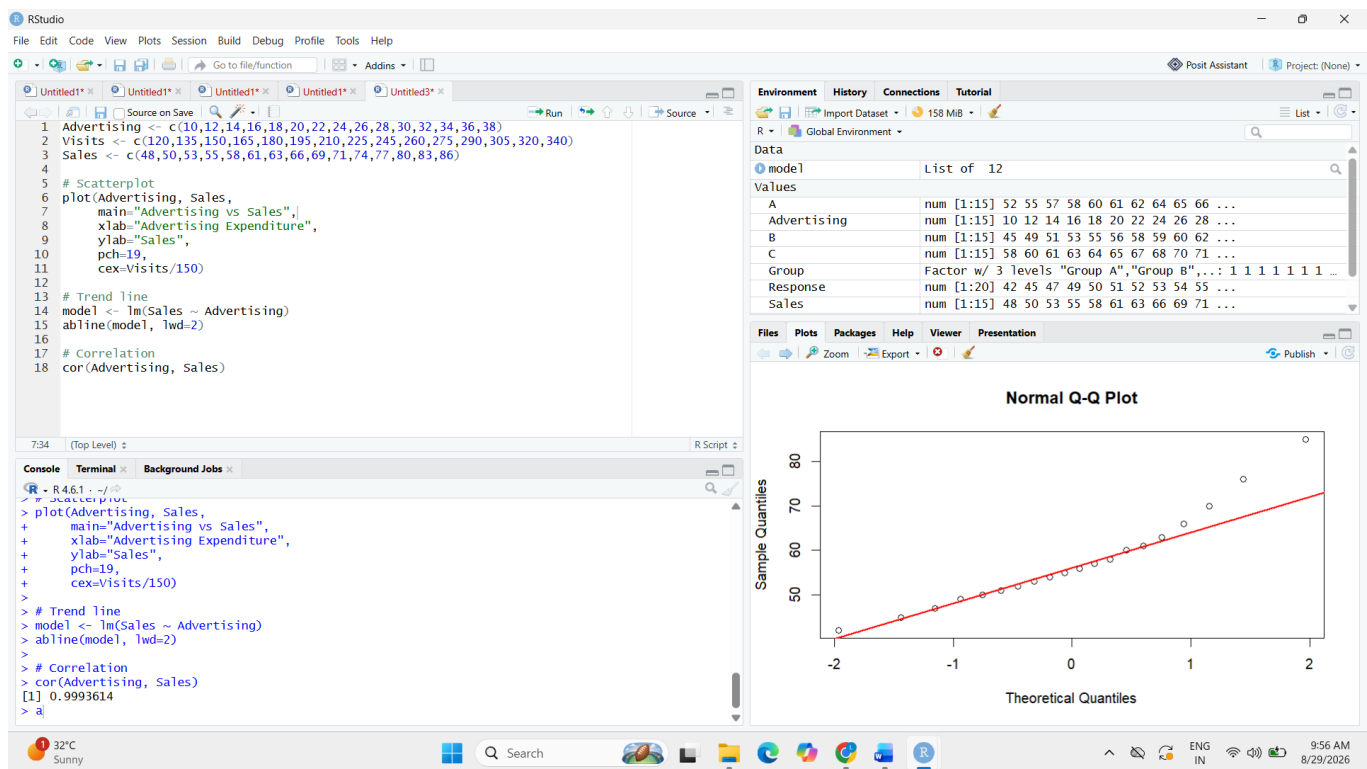
```
qqline(Response, col="red", lwd=2)
```

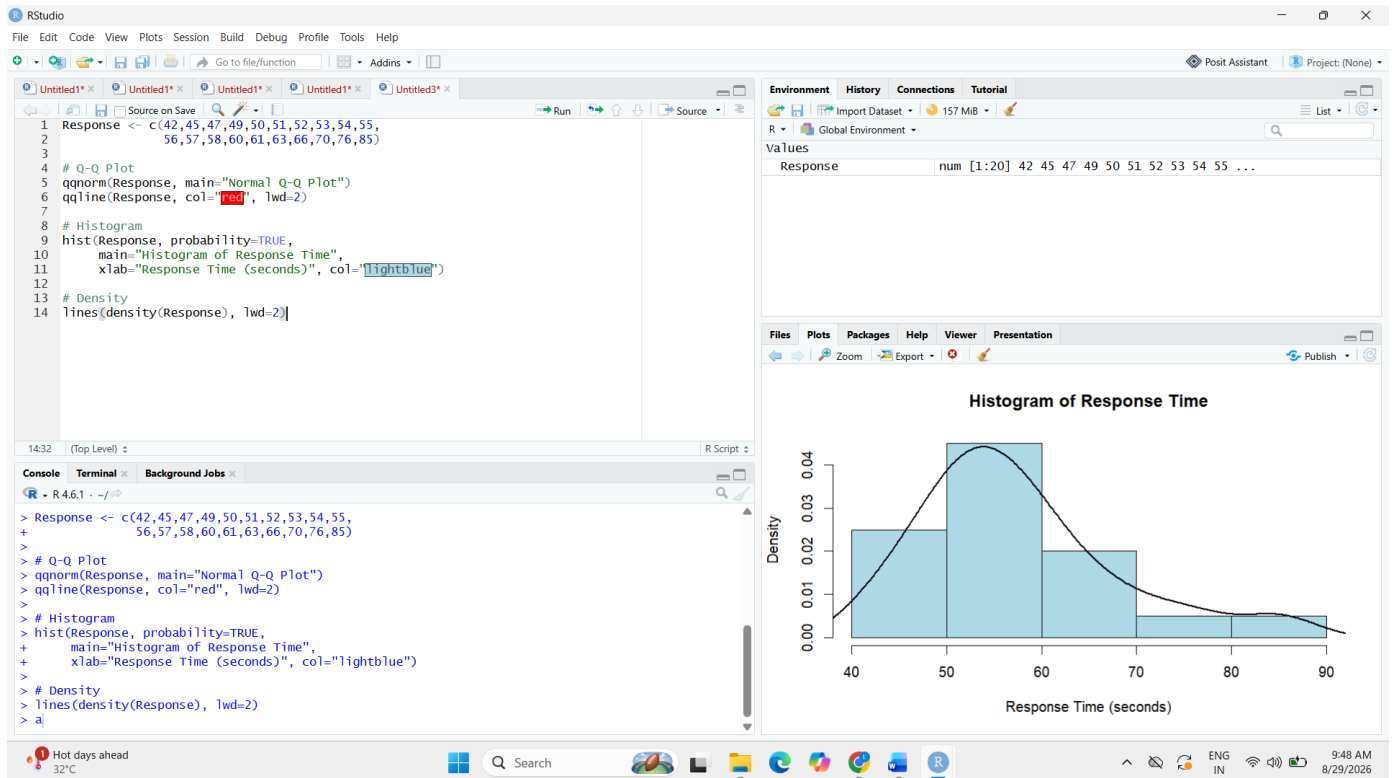
```
# Histogram
```

```
hist(Response, probability=TRUE,  
      main="Histogram of Response Time",  
      xlab="Response Time (seconds)", col="lightblue")
```

```
# Density
```

```
lines(density(Response), lwd=2)
```





Result / Interpretation

- Most observations lie reasonably close to the Q-Q reference line.
- The upper observations **76 and 85** deviate noticeably from the line.
- This indicates a **right-tail departure** and possible high-value outliers.
- Therefore, the data is **approximately normal but not perfectly normal**.
- The histogram also shows a slight **right skew**.

Q3. Visualizing Many Distributions at Once

Aim

To compare the center, spread, and outliers of three groups of examination scores using grouped boxplots.

R Code

```
A <- c(52,55,57,58,60,61,62,64,65,66,68,70,72,74,78)
```

```
B <- c(45,49,51,53,55,56,58,59,60,62,63,65,67,69,71)
```

```
C <- c(58,60,61,63,64,65,67,68,70,71,72,74,76,80,84)
```

```
Scores <- c(A, B, C)
```

```
Group <- factor(rep(c("Group A","Group B","Group C"), each=15))
```

```
boxplot(Scores ~ Group,
```

```
  main="Comparison of Examination Scores",
```

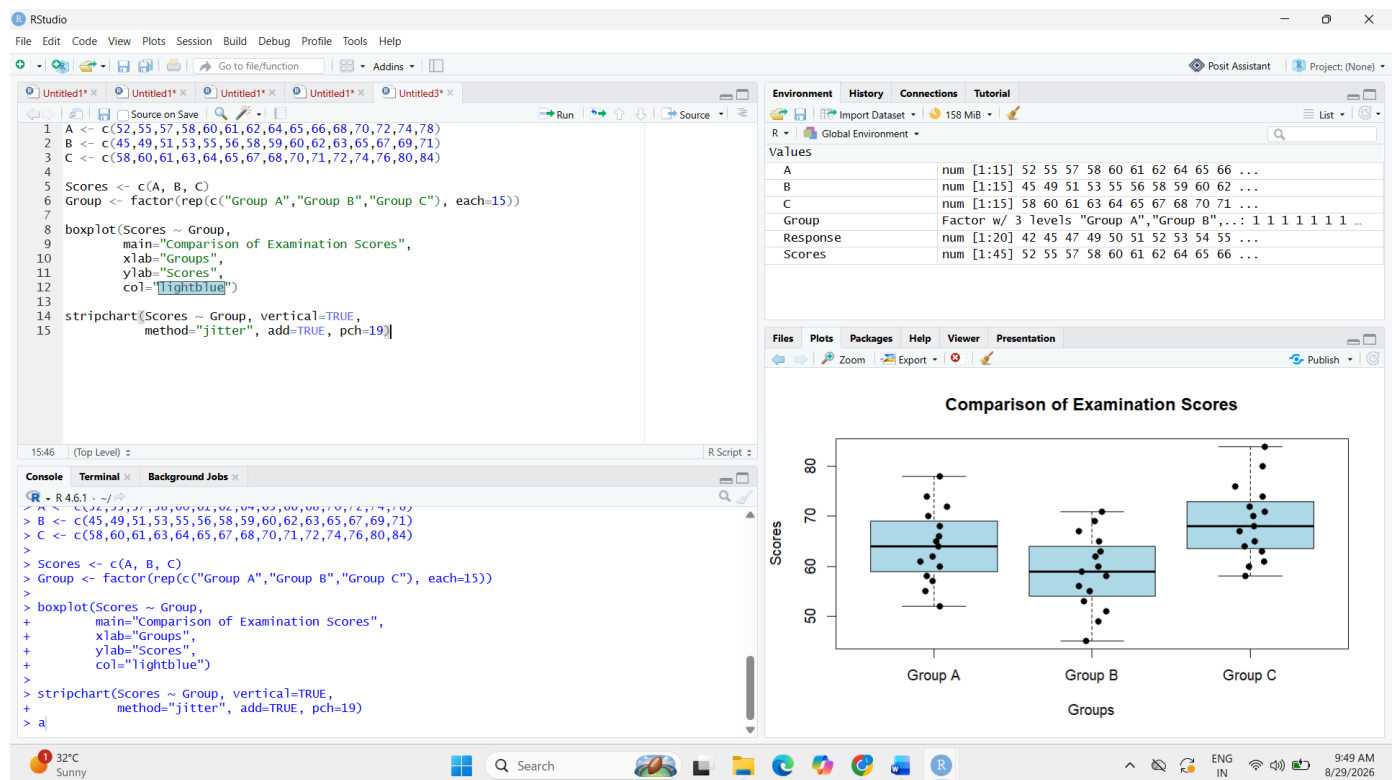
```
  xlab="Groups",
```

```
  ylab="Scores",
```

```
  col="lightblue")
```

```
stripchart(Scores ~ Group, vertical=TRUE,
```

```
  method="jitter", add=TRUE, pch=19)
```



Result / Interpretation

- **Group A:** Center is around the mid-60s with moderate spread.
- **Group B:** Has the **lowest center** and includes lower scores such as 45.
- **Group C:** Has the **highest center** and greater upper spread, reaching 84.

- There are no major extreme outliers.
 - A grouped boxplot is better than three separate plots because all groups can be compared **side-by-side using the same scale**.
-

Q4. Relationships and Multivariate Visualization

Aim

To analyze the relationship between advertising expenditure and sales while incorporating website visits as an additional variable.

R Code

```
Advertising <- c(10,12,14,16,18,20,22,24,26,28,30,32,34,36,38)
```

```
Visits <- c(120,135,150,165,180,195,210,225,245,260,275,290,305,320,340)
```

```
Sales <- c(48,50,53,55,58,61,63,66,69,71,74,77,80,83,86)
```

```
# Scatterplot
```

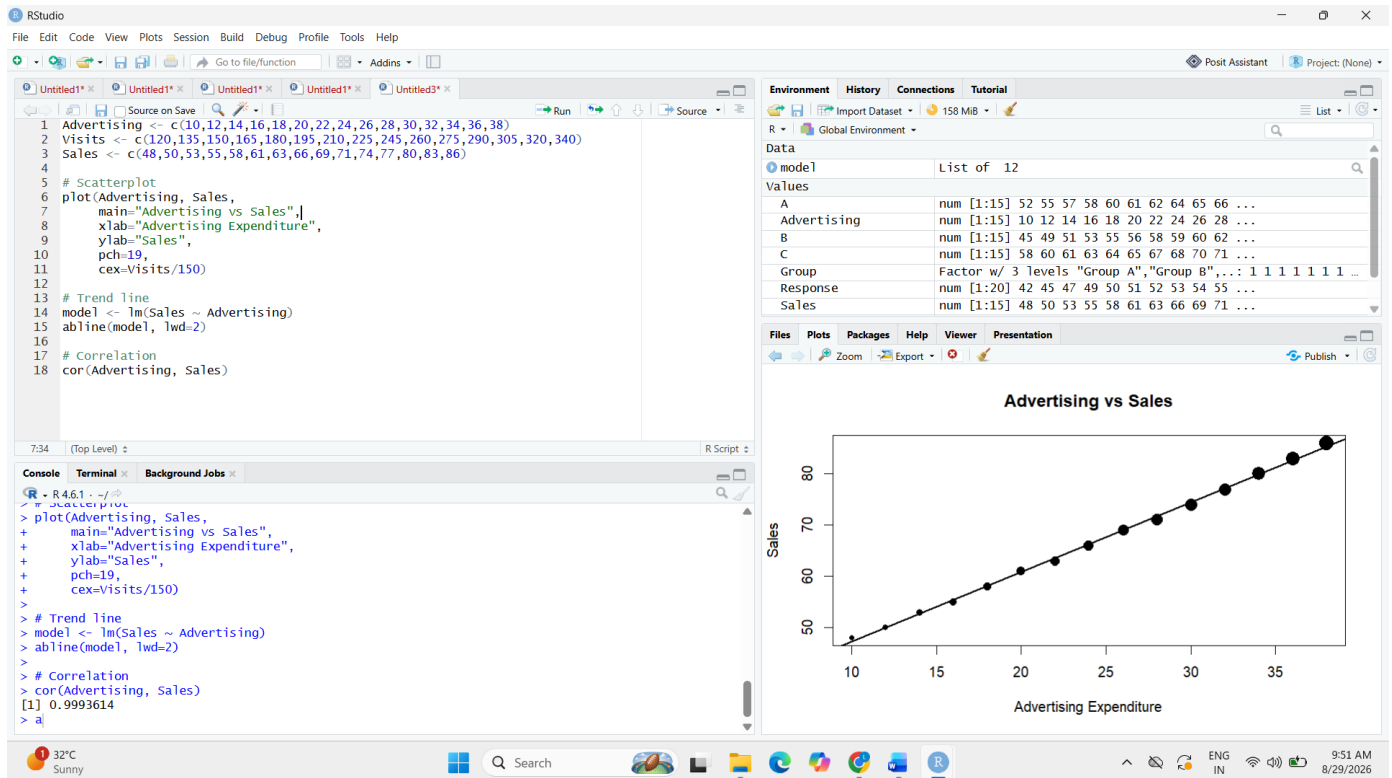
```
plot(Advertising, Sales,  
     main="Advertising vs Sales",  
     xlab="Advertising Expenditure",  
     ylab="Sales",  
     pch=19,  
     cex=Visits/150)
```

```
# Trend line
```

```
model <- lm(Sales ~ Advertising)  
abline(model, lwd=2)
```

```
# Correlation
```

```
cor(Advertising, Sales)
```

Result / Interpretation

- There is a **strong positive relationship** between advertising expenditure and sales.
- As advertising expenditure increases, sales also increase.
- Website visits can be represented by **point size**.
- This gives additional information without creating another graph.
- However, too many visual encodings can make the graph cluttered.
- The trend line clearly shows the increasing relationship.

Correlation ≈ 0.998 , indicating a very strong positive relationship.

Q5. Misleading Visualization and Redesign

Aim

To demonstrate how a truncated y-axis can exaggerate differences and redesign the visualization using an honest baseline.

R Code

```
Department <- c("CSE","ECE","EEE","MECH","CIVIL","AIDS","IT")
```

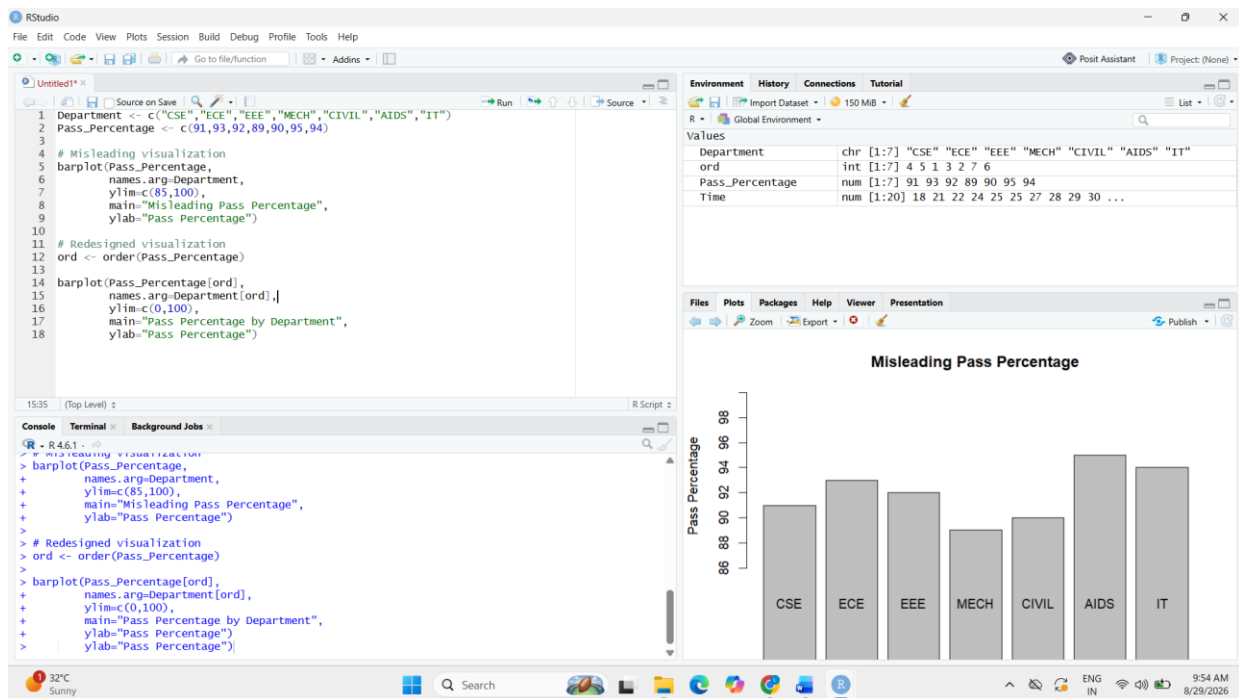
```
Pass_Percentage <- c(91,93,92,89,90,95,94)
```

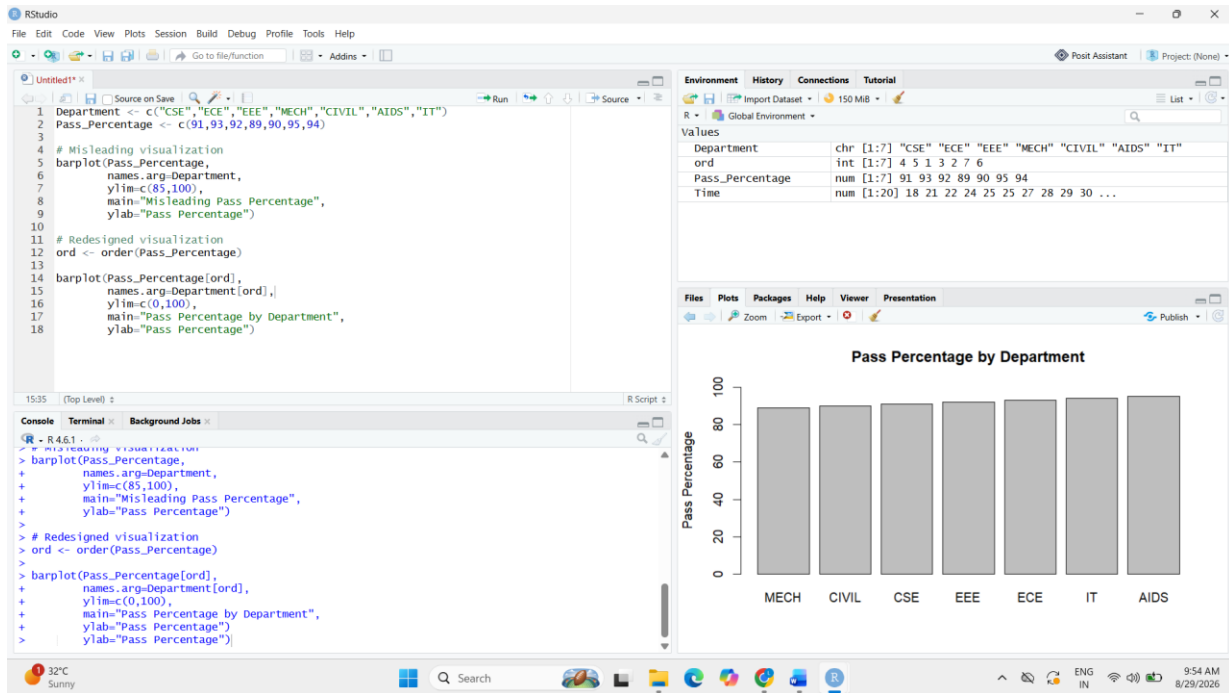
Misleading visualization

```
barplot(Pass_Percentage,  
        names.arg=Department,  
        ylim=c(85,100),  
        main="Misleading Pass Percentage",  
        ylab="Pass Percentage")
```

Redesigned visualization

```
ord <- order(Pass_Percentage)  
  
barplot(Pass_Percentage[ord],  
        names.arg=Department[ord],  
        ylim=c(0,100),  
        main="Pass Percentage by Department",  
        ylab="Pass Percentage")
```





Result / Interpretation

- The first graph uses a **truncated y-axis starting at 85%**.
- This makes small differences such as 89% vs 95% appear much larger than they really are.
- The second graph starts at **0**, giving an honest visual comparison.
- Departments are arranged from lowest to highest pass percentage.
- Clear labels and an appropriate baseline improve accuracy and readability.
- Therefore, the redesigned graph communicates the data **more honestly and effectively**.